

RAJ MOHAN

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SUMMARY

Backend Engineer with hands-on experience developing scalable **RESTful APIs** and production-grade web applications using **Python, Django, and FastAPI**. Proficient in **JWT authentication**, asynchronous task pipelines with **Celery and Redis**, real-time systems via **WebSockets**, and **CI/CD** workflows. Skilled in **PostgreSQL, Docker, Linux**, and cloud deployment on **AWS**. Proven delivery through academic projects and state-level hackathon wins.

EDUCATION

SJB Institute of Technology

Bachelor of Engineering in Information Science & Engineering | CGPA: 8.04 / 10.00

Sep. 2023 – May 2027

Bengaluru, India

RELEVANT COURSEWORK

- Object-Oriented Programming (OOP)
- Database Management Systems
- Operating Systems
- Computer Networks

EXPERIENCE

MythoQuantum Explorers

Research Analyst Intern

Nov 2024 – Jan 2025

Remote

- Conducted structured data research and exploratory analysis across **5+ projects**, synthesizing findings into actionable insights that supported team decision-making and accelerated project delivery.
- Collaborated cross-functionally in an **Agile** environment, producing concise research reports and identifying data patterns that improved research clarity and contributed to data-driven outcomes.

PROJECTS

Online Exam Integrity Monitor – Real-Time Proctoring System 🐙 [repo](#)

Python, Django, Django Channels, WebSockets, Celery, Redis, PostgreSQL, ReportLab, REST API

- Built a real-time exam proctoring system supporting **50+ concurrent sessions** using **Django Channels and WebSockets**, achieving sub-second event propagation and live session monitoring across all active users.
- Designed a **Celery + Redis** async pipeline processing **100+ violation events per session**, with **Celery Beat** scheduling automated PDF report generation via ReportLab for examiner review.
- Implemented **JWT-based role-based access control** (Student / Examiner), RESTful API endpoints, webcam snapshot capture via MediaDevices API, and a real-time dashboard with dark-mode UI.

Travel Itinerary Planner – Web-Based Trip Planning Platform 🐙 [repo](#)

Python, Django, REST API, JSON, HTML, CSS, OpenWeatherMap API, Google Maps API

- Developed a full-stack trip planning application using **Django**, integrating **3 third-party REST APIs** – Google Maps for route optimization, OpenWeatherMap for live forecasts – serving personalized itineraries to users.
- Engineered a modular backend with **MVC architecture**, **JSON** data exchange, robust **HTTP** error handling, and clean ORM-based **PostgreSQL** queries, reducing API failure rate by **~40%**.

TECHNICAL SKILLS

Languages: Python, C++, SQL, JSON, HTML, CSS

Frameworks & Libraries: Django, Django REST Framework, Django Channels, FastAPI, Celery

Concepts: RESTful API, Microservices, JWT Authentication, ORM, CI/CD, Agile, Scrum, WebSockets

Databases & Brokers: PostgreSQL, Redis, SQLite

Tools & Cloud: Git, GitHub Actions, Docker, AWS (EC2, S3), Linux, VS Code

CERTIFICATIONS & ACHIEVEMENTS

- **AWS Cloud Technical Essentials** – Amazon Web Services (Coursera), Apr 2026 | [View Certificate](#)
- **2× State-Level Hackathon Winner**